NOT MEASUREMENT SENSITIVE

MIL-S-20633C
16 February 1989
SUPERSEDING
MIL-S-20633B
19 October 1965

MILITARY SPECIFICATION

SIGNS AND MARKER, ALUMINUM;

MINED AREA MARKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers safe-lane and mined-area marking signs and a mine marker.

1.2 Classification. Signs and markers shall be of the following classes, as specified (see 6.2, 6.4 and 6.5).

Class D - safe-lane sign.
Class E - mine marker.
Class F - mined-area sign.

1.3 Use of international standardization agreement code numbers. This specification is the subject of international standardization agreement, the NATO, STANAG 2001 and 2002 (see 6.7).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

UU-P-268	- Paper, Kraft, Untreated Wrapping.
PPP-B-636	- Box, Fiberboard.

MILITARY

MIL-P-116	- Preservations, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Drawings. The following drawings, form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

13211E7810	- Sign, Mined Area Marking.
13211E7811	- Sign, Safe Lane Marking.
13211E7812	- Marker, Land Mine.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-ND, Fort Belvoir, VA 22060-5606.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Design and construction. The class D safe-lane sign shall be fabricated in accordance with Drawing 13211E7811. The class E mine marker shall be fabricated in accordance with Drawing 13211E7812. The class F mined-area sign shall be fabricated in accordance with Drawing 13211E7810.

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3.3 Drawings. The drawings forming a part of this specification are end product drawings. The contractor is responsible for preparing his own shop drawings. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained rom a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer. The contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the signs and marker. No deviation from the prescribed dimensions or tolerances is permissible without prior approval from the contracting officer.

3.4 Material. Materials shall be as specified herein and on the applicable drawings. The materials shall be free from all defects and imperfections that my affect the quality and reliability of the finished product.

3.4.1 Deterioration prevention and control. The signs and markers shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the signs and markers may be exposed.

3.4.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.5 Treatment and painting. The portions of the signs and markers normally painted shall be cleaned, treated, and painted in accordance with requirements specified on the applicable drawings.

3.6 Workmanship. All signs and markers shall be fabricated and finished to ensure freedom from sharp edges, burrs, blemishes and defects; accuracy of dimensions; thoroughness of riveting; and alignment of parts where applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the

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contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.5).

4.3 First article inspection. When specified (see 6.2), a random sample shall be subject to first article inspection (see 6.3). The contractor shall furnish to the Government one or more safe-lane and mined-area marking signs and mine markers, as specified by the contract or the contracting officer, produced by production tooling for inspection. Examination and tests shall be as specified in section 4; and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subjected to surveillance and approval by the Government. The signs and markers shall be examined for the characteristics in column 2 of table I of 4.4.3.1 to determine conformance to the requirements of this specification. Acceptance of first article signs and markers shall not exclude the remaining signs and markers from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the first article test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the first article test (see 6.3), and failure of such additional tests shall have the same effect as failure of those tests specified as first article tests.

4.3.1 Failure of inspection. Failure of first article signs and markers to meet examination criteria in column 2 of table I of 4.4.3.1 shall be cause for rejection of the first article signs and markers. Further, any failure shall be cause for the Government to refuse to accept any production signs and markers until the contractor has provided acceptable evidence that action has been taken to eliminate the defects. Correction of such defects shall be accomplished by the contractor at no cost to the Government on signs and markers previously accepted and produced under the contract. Defects found as a result of the first article inspection will be considered prima facie evidence that all signs and markers accepted prior to the completion of first article inspection are similarly defective unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Unit of product. The unit of product is one safe-lane sign, one mined-area sign, and one mine marker.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Lot size will be specified by the contracting officer.

4.4.3 Inspection procedures.

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4.4.3.1 Examination. Samples selected in accordance with 4.4.2 shall be examined for the defects in column 2 of table I. AQL shall be 1.0 percent defective.

TABLE I. Examination.

Number	Characteristics	Requirement
101.	Components missing or not assembled as specified.	3.2
102.	Dimensions not as specified.	3.3
103.	Materials not as specified.	3.4, 3.4.1
104.	Treatment and paint not as specified.	3.5
105.	Workmanship not as specified.	3.6

4.5 Inspection of packaging.

4.5.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Lot size will be specified by the contracting officer.

4.5.3 Examination. Samples selected in accordance with 4.5.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

106. Preservation not as specified for level A or C.
107. Unit pack quantity not as specified for level A or C.
108. Packing not as specified for level A or C.
109. Strapping not finish B for level A.
110. Marking not as specified for all levels.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. All signs and markers of like description, class D, E, and F shall be in quantities of 10 in accordance with MIL-P-116, method IC-3. The signs and markers shall be interleaved with a single sheet of paper conforming to UU-P-268, type I.

5.1.2 Level C. All signs and markers, Class D, E, and F, shall be preserved in accordance with MIL-P-116, method III.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. All signs and markers of the same class, preserved as specified in 5.1, shall be packed in boxes conforming to PPP-B-636, weather resistant type, style optional. The boxes shall be close fitting and strapped in accordance with the appendix of the box specification. The strapping shall be flat steel, finish B.

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5.2.2 Level C. All signs and markers of the same class, preserved as specified in 5.1, shall be closely packed in boxes conforming to PPP-B-636, domestic type, style optional. Boxes shall be closed and sealed in accordance with the box specification.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), all packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The class D sign is intended for marking a safe lane through a mined area. The class E mine marker is intended for marking the location of a detected mine until removal is accomplished. The class F sign is intended for marking mined areas.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Class required (see 1.2).
d. Time frame required for submission of first article model(s) and number of signs and markers required (see 3.1).
e. When the Government will conduct any or all of the first article production model examinations and tests. When the Government will conduct some but not all of the first article examinations and tests, the contracting officer should specify which examinations and tests shall be conducted by the contractor (see 3.1 and 4.3).
f. When the contractor is required to inspect a quantity of first article models other than stated in 4.4, the contracting officer should specify the quantity the contractor is to inspect.
g. Level of preservation and packing required (see 5.1 and 5.2).
h. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Class A, B, C. Class A, B, and C signs have been discontinued since they are no longer required.

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6.5 Part or identifying number (PIN). The PINs to be used for items acquired in this specification are created as follows:

M	20633	D, E or F
Prefix to indicate military specification	specification number	class letter

6.6 Subject term (key word) listing.

Marker, mine
Sign, mined-area
Sign, safe-lane

6.7 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement, the NATO, STANAG 2001 and 2002. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activities:
Army - MI
DLA - GS
Air Force - 84

User activity:
Army - GL

Preparing activity:
Army - ME

Project 9905-0290

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-S-20633C	Signs and Marker, Aluminum; Mined Area Marking

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*

5a. Recommended Wording

5b. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8 DATE OF SUBMISSION *(YYMMDD)*

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